

%nprocshared=64
%mem=320GB
%chk=PbRhaC8C8T2.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC8C8T2

0 1

Pb	-0.26274800	1.20737400	-0.87859800
O	1.39563100	-0.61839100	1.69393600
O	0.67766400	2.12742600	-2.92718500
O	-0.35889000	-1.27659600	-0.13438200
C	1.80917800	-4.29492000	0.36252200
O	2.99558100	-3.49633600	0.23012000
C	3.00626000	-2.34685400	1.05019300
C	1.93466400	-1.33292100	0.58462200
C	0.81532700	-2.08503700	-0.12449800
C	0.56979200	-3.41491700	0.56896400
O	1.51939300	0.24463900	-2.14323900
C	1.55607400	1.22074400	-2.95221200
C	2.72154100	1.31233900	-3.90533500
C	3.87159300	2.11491500	-3.27928400
O	4.13898800	1.63824100	-1.93912900
C	4.75507800	0.46030600	-1.81442900
O	5.15197300	-0.19785500	-2.74962800
C	4.88269000	0.06087300	-0.36767400
C	4.84285200	-1.46480600	-0.20738500
O	4.29210000	-1.81510100	1.07478000
C	1.95755000	-5.33367900	1.45700000
O	-0.58132200	-4.05996700	0.05777100
O	-0.96857300	-0.59047500	4.46215900
O	-2.82409500	1.69492000	-1.78311100
O	0.80937800	-2.67870100	3.72210400
C	-2.03531300	-4.39450000	3.48089800
O	-1.90588600	-3.44662000	2.41936200
C	-1.91144400	-2.09620400	2.83115300
C	-1.43237100	-1.91324300	4.26944700
C	-0.30514200	-2.87306800	4.57551400
C	-0.81952000	-4.31913400	4.41332000
O	-1.59862000	-0.09098800	-2.24846200
C	-2.69534900	0.57001700	-2.29902500
C	-3.85348400	-0.09032500	-3.01822000
C	-5.15117800	0.02822200	-2.24542200
O	-4.91559300	-0.62709000	-0.97010200
C	-5.37153100	-0.04168700	0.14297900
O	-5.99123700	0.99564200	0.17113700
C	-4.97349700	-0.84030500	1.35857500
C	-3.45721300	-0.87463100	1.51402000
O	-3.19004200	-1.53398300	2.75560300
C	-2.20216400	-5.75854100	2.85882600

O	-1.25996800	-4.86951800	5.64063700
H	-3.01801510	-1.42184870	0.70620938
H	-5.39596091	1.05836937	-2.09122174
H	4.23547231	-1.88640743	-0.98084615
H	4.75084164	2.00043045	-3.87821231
H	1.70139600	-4.78406200	-0.61078400
H	2.78800200	-2.61458600	2.09086100
H	2.36832700	-0.62739000	-0.12556500
H	1.13478100	-2.26188100	-1.15683100
H	0.43026500	-3.23526400	1.63643100
H	3.06638800	0.30574100	-4.13943700
H	2.41444400	1.81335500	-4.82474500
H	5.82902700	0.46341900	0.01189400
H	4.09200300	0.54430200	0.20547800
H	1.05300800	-5.94494000	1.50792600
H	2.09785000	-4.85615500	2.42991900
H	2.80677900	-5.99061000	1.25808800
H	-1.25597200	-3.96201800	0.74905500
H	-2.92662300	-4.14695800	4.07151700
H	-1.22818000	-1.58245100	2.15027800
H	-2.26283000	-2.13209900	4.95282200
H	0.00042600	-2.73425300	5.62104000
H	0.00221100	-4.90985100	3.99026000
H	-3.97508600	0.40824700	-3.98541400
H	-3.61366300	-1.13735300	-3.21263100
H	-5.43401700	-0.38360700	2.23465400
H	-5.34860900	-1.86205600	1.25904300
H	-2.25896700	-6.51625600	3.64238500
H	-1.34989800	-5.99141900	2.21536400
H	-3.11490300	-5.80590800	2.26149800
H	-0.49264700	-5.03616800	6.19723400
H	-1.69114400	0.01257400	4.25390000
H	0.85853400	-1.75088000	3.44876100
H	-0.96988600	-1.65859700	-0.77943700
H	2.06874500	-0.00597300	2.00960300
C	3.48299491	3.60401688	-3.22300682
H	2.88072901	3.84783496	-4.07314124
H	2.92950674	3.79458572	-2.32733145
C	4.75781867	4.46792571	-3.23227451
H	5.16686995	4.51069933	-2.24447521
H	5.47628839	4.03637554	-3.89745547
C	4.40985469	5.89138172	-3.70587146
H	4.03982669	5.85300649	-4.70911950
H	3.66067827	6.30731632	-3.06506487
C	5.67270840	6.77143393	-3.65782348
H	6.43250761	6.33895008	-4.27472128
H	6.02514519	6.83477551	-2.64951988
C	5.33477334	8.18277465	-4.17310717
H	6.21173595	8.79481437	-4.13783730
H	4.98423706	8.11964126	-5.18208611

H	4.57354804	8.61447476	-3.55741971
C	6.27243948	-2.02649148	-0.31869111
H	6.38976922	-2.51895362	-1.26135477
H	6.97734723	-1.22489553	-0.24487381
C	6.51861914	-3.03309832	0.82049204
H	6.27729613	-2.57697806	1.75783804
H	5.90199942	-3.89521980	0.67411349
C	7.99959481	-3.45536654	0.82127295
H	8.25857954	-3.84958539	-0.13915321
H	8.61283726	-2.60496617	1.03493838
C	8.22521487	-4.53419483	1.89683099
H	7.68098880	-5.41762727	1.63555507
H	7.88285775	-4.17124463	2.84338198
C	9.72685172	-4.86391733	1.98610490
H	9.87939682	-5.63988159	2.70687495
H	10.07756170	-5.19147851	1.02975445
H	10.26694131	-3.98948466	2.28371327
C	-2.89735422	0.55962156	1.54684876
H	-1.83769170	0.52564099	1.69128096
H	-3.11556923	1.04896453	0.62065997
C	-3.54855101	1.33849216	2.70482257
H	-4.10966696	2.16005359	2.31101334
H	-4.20170312	0.68868523	3.24892115
C	-2.45210942	1.87016431	3.64649160
H	-2.42902886	2.93888284	3.59950248
H	-1.50275245	1.47931423	3.34506572
C	-2.75525124	1.42621243	5.08961729
H	-3.34318110	2.17552204	5.57723764
H	-3.29643753	0.50332079	5.07263422
C	-1.43330993	1.23208904	5.85537657
H	-1.44050889	0.28006821	6.34374580
H	-0.61381161	1.27450743	5.16870110
H	-1.32711798	2.00681364	6.58573612
C	-6.28802560	-0.68076171	-3.00471089
H	-6.35716176	-0.28328344	-3.99573623
H	-6.08356067	-1.72985645	-3.05465412
C	-7.61964494	-0.45190468	-2.26578303
H	-7.67062669	-1.09783917	-1.41427291
H	-7.67876247	0.56702945	-1.94456914
C	-8.79280314	-0.75988835	-3.21470132
H	-8.79771860	-0.05213137	-4.01716994
H	-8.68124031	-1.74755087	-3.61090390
C	-10.11945328	-0.66501726	-2.43842889
H	-10.21217168	0.31042561	-2.00852832
H	-10.13109163	-1.40003084	-1.66092032
C	-11.29590712	-0.91618468	-3.39992318
H	-12.21673378	-0.86787612	-2.85711148
H	-11.29474541	-0.17058860	-4.16737685
H	-11.19364807	-1.88460171	-3.84333217

C H O 0

6-311+g(2d,p)

Pb 0

S 4 1.00

591.61124370 0.22126521076D-03

46.757232559 0.56961959130D-02

20.746462696 -0.21374063831

14.610796419 0.40502620616

S 2 1.00

20.222637612 -0.83541883299D-01

6.4767324865 0.97910892388

S 1 1.00

1.6600600927 1.0000000

S 1 1.00

0.80431655001 1.0000000

S 1 1.00

0.22627039020 1.0000000

S 1 1.00

0.84014530665D-01 1.0000000

P 3 1.00

15.189102118 0.61952303583

14.693144415 -0.72498497086

6.8705890048 0.37680007984

P 3 1.00

2.2021426123 0.40196284806

1.2209125119 0.46058131862

0.63367559815 0.19367655397

P 1 1.00

0.28202837058 1.0000000

P 1 1.00

0.11333375666 1.0000000

P 1 1.00

0.43948707430D-01 1.0000000

D 6 1.00

61.315369628 0.33870800787D-03

12.372195840 0.13788683942D-01

6.9254944983 -0.75979608103D-01

2.3319539939 0.28113784298

1.2108730003 0.44474512269

0.60090478506 0.35326874351

D 1 1.00

0.28135869813 1.0000000

D 1 1.00

0.11500000000 1.0000000

F 1 1.00

1.0000000 1.0000000

F 1 1.00

0.2896200 1.0000000

PB	0	
PB-ECP	3	60
f potential		
2		
2	3.88751200	12.20989200
2	3.81196300	16.19029100
s-f potential		
4		
2	12.29630300	281.28549900
2	8.63263400	62.52021700
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
p-f potential		
6		
2	10.24179000	72.27689700
2	8.92417600	144.59108300
2	6.58134200	4.75869300
2	6.25540300	9.94062100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100
d-f potential		
6		
2	7.75433600	35.84850700
2	7.72028100	53.72434200
2	4.97026400	10.11525600
2	4.56378900	14.83373100
2	3.88751200	-12.20989200
2	3.81196300	-16.19029100